

LOGICAL DEDUCTION IN AI

INFERRING BY RESOLUTION REFUTATION



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Predicate Logic

Wherever Mary goes, so does the lamb. Mary goes to school. So the lamb goes to school.

No contractors are dependable. Some engineers are contractors. Therefore some engineers are not dependable.

All dancers are graceful. Ayesha is a student. Ayesha is a dancer. Therefore some student is graceful.

Every passenger is either in first class or second class. Each passenger is in second class if and only if he or she is not wealthy. Some passengers are wealthy. Not all passengers are wealthy. Therefore some passengers are in second class.

New Additions in Proposition (First Order Logic)

Variables, Constants, Predicate Symbols and New Connectors: $\exists$ (there exists), $\forall$ (for all)

Wherever Mary goes, so does the Lamb. Mary goes to School. So the Lamb goes to School.

Predicate: $\text{goes}(x,y)$ to represent x goes to y

New Connectors: $\exists$ (there exists), $\forall$ (for all)

F1: $\forall x(\text{goes}(\text{Mary}, x) \rightarrow \text{goes}(\text{Lamb}, x))$

F2: $\text{goes}(\text{Mary}, \text{School})$

G: $\text{goes}(\text{Lamb}, \text{School})$

To prove: $(F1 \wedge F2) \rightarrow G$ is always true

Inferencing in Predicate Logic

Domain: **D**

Constant Symbols: **M, N, O, P, ...**

Variable Symbols: **x, y, z, ...**

Function Symbols: **F(x), G(x, y),
H(x, y, z)**

Predicate Symbols: **p(x), q(x, y),
r(x, y, z),**

Connectors: **$\sim, \wedge, \vee, \rightarrow, \exists, \forall$**

Terms:

Well-formed Formula:

Free and Bound Variables:

**Interpretation, Valid, Non-Valid,
Satisfiable, Unsatisfiable**

What is an Interpretation? Assign a domain set D, map constants, functions, predicates suitably. **The formula will now have a truth value**

Example:

F1: $\forall x(g(M, x) \rightarrow g(L, x))$

F2: $g(M, S)$

G: $g(L, S)$

Interpretation 1: **D = {Akash, Baby, Home, Play, Ratan, Swim}, etc.,**

Interpretation 2: **D = Set of Integers, etc.,**

How many interpretations can there be?

To prove Validity, means $(F1 \wedge F2) \rightarrow G$ is true under all interpretations

To prove Satisfiability means $(F1 \wedge F2) \rightarrow G$ is true under at least one interpretation

Resolution Refutation for Propositional Logic

To prove validity of

$F = ((F1 \wedge F2 \wedge \dots \wedge F_n) \rightarrow G)$

we shall attempt to prove that

$\sim F = (F1 \wedge F2 \wedge \dots \wedge F_n \wedge \sim G)$

is unsatisfiable

Steps for Proof by Resolution Refutation:

1. Convert of Clausal Form / Conjunctive Normal Form (CNF, Product of Sums).
2. Generate new clauses using the resolution rule.
3. At the end, either False will be derived if the formula $\sim F$ is unsatisfiable implying F is valid.

If Asha is elected VP then Rajat is chosen as G-Sec and Bharati is chosen as Treasurer. Rajat is not chosen as G-Sec. Therefore Asha is not elected VP.

$F1: (a \rightarrow (b \wedge c)) = (\sim a \vee b) \wedge (\sim a \vee c)$

$F2: \sim b, G: \sim a, \sim G: a$

Clauses of Clause Form: $\sim F = (C1 \wedge C2 \wedge C3 \wedge C4)$

where: $C1: (\sim a \vee b)$

$C2: (\sim a \vee c)$

$C3: \sim b$

$C4: a$

To prove that $\sim F$ is False

Let $C1 = a \vee b$ and $C2 = \sim a \vee c$

then a new clause $C3 = b \vee c$ can be derived.

(Proof by showing that $((C1 \wedge C2) \rightarrow C3)$ is a valid formula).

To prove unsatisfiability use the Resolution Rule repeatedly to reach a situation where we have two contradictory clauses of the form $C1 = a$ and $C2 = \sim a$ from which **False** can be derived.

If the propositional formula is satisfiable then we will not reach a contradiction and eventually no new clauses will be derivable.

For propositional logic the procedure terminates.

Resolution Rule is **Sound** and **Complete**

Applying Resolution Refutation

Let $C1 = a \vee b$ and $C2 = \sim a \vee c$
then a new clause $C3 = b \vee c$ can be derived.

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F2: $\sim b$

G: $\sim a$

$\sim G:$ a

Clauses of Clause Form: $\sim F$
 $= (C1 \wedge C2 \wedge C3 \wedge C4)$

where: $C1: (\sim a \vee b)$

$C2: (\sim a \vee c)$

$C3: \sim b$

$C4: a$

To prove that $\sim F$ is False

New Clauses Derived

C5: $\sim a$ (Using C1 and C3)

C6: **False** (using C4 and C5)

Solve Using Resolution Refutation

Suppose we know that: “if Arjun is thin, then Mohit is not bearded or Julia is not tall” and “if Julia is tall then Devika is graceful” and “if Devika is graceful and Mohit is bearded then Arjun is thin” and “Mohit is bearded”. Can we deduce that “Julia is not tall” ?

Example

Let $C1 = a \vee b$ and $C2 = \sim a \vee c$
then a new clause $C3 = b \vee c$ can
be derived.

(Proof by showing that $((C1 \wedge C2) \rightarrow C3)$ is a valid formula).

To prove unsatisfiability use the
Resolution Rule repeatedly to
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two contradictory clauses of the
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If the propositional formula is
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For propositional logic the
procedure terminates.

Resolution Rule is **Sound** and
Complete

Rajesh either took the bus or came by cycle to class. If he came by cycle or walked to class he arrived late. Rajesh did not arrive late. Therefore he took the bus to class.

Resolution Refutation for Predicate Logic

Given a formula F which we wish to check for validity, we first check if there are any free variables. We then quantify all free variables universally.

Create $F' = \sim F$ and check for unsatisfiability of F'

STEPS:

Conversion to Clausal (CNF) Form:

- Handling of Variables and Quantifiers, Ground Instances

Applying the Resolution Rule:

- Concept of Unification
- Principle of Most General Unifier (mgu)
- Repeated application of Resolution Rule using mgu

CONVERSION TO CLAUSAL FORM IN PREDICATE LOGIC

1. Remove implications and other Boolean symbols converting to equivalent forms using $\sim$, $\vee$, $\wedge$
2. Move negates ($\sim$) inwards as close as possible
3. Standardize (Rename) variables to make them unambiguous
4. Remove Existential Quantifiers by an appropriate new function /constant symbol taking into account the variables dependent on the quantifier (Skolemization)
5. Drop Universal Quantifiers
6. Distribute $\vee$ over $\wedge$ and convert to CNF

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F2: $\text{goes}(\text{Mary}, \text{School})$

G: $\text{goes}(\text{Lamb}, \text{School})$

To prove: $(F1 \wedge F2) \rightarrow G$ is valid

CONVERSION TO CLAUSAL FORM IN PREDICATE LOGIC

1. Remove implications and other Boolean symbols converting to equivalent forms using $\sim, \vee, \wedge$
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6. Distribute $\vee$ over $\wedge$ and convert to CNF

Conversion to Clausal Form

1. Remove implications and other Boolean symbols converting to equivalent forms using $\sim$, $\vee$, $\wedge$
2. Move negates ($\sim$) inwards as close as possible
3. Standardize (Rename) variables to make them unambiguous
4. Remove Existential Quantifiers by an appropriate new function /constant symbol taking into account the variables dependent on the quantifier (Skolemization)
5. Drop Universal Quantifiers
6. Distribute $\vee$ over $\wedge$ and convert to CNF

$\forall x(\forall y(\text{student}(y) \rightarrow \text{likes}(x, y)) \rightarrow (\exists z(\text{likes}(z, x))))$

Substitution, Unification, Resolution

Consider clauses:

- C1: $\sim \text{studies}(x,y) \vee \text{passes}(x,y)$
- C2: $\text{studies}(\text{Madan},z)$
- C3: $\sim \text{passes}(\text{Chetan}, \text{Physics})$
- C4: $\sim \text{passes}(w, \text{Mechanics})$

What new clauses can we derive by the resolution principle?

Ground Clause and a more general clause

Concept of substitution / unification and the Most General Unifier (mgu)

Resolution Rule for Predicate Calculus: Repeated Application of Resolution using mgu

Examples

F1: $\forall x(\text{contractor}(x) \rightarrow \sim \text{dependable}(x))$

F2: $\exists x(\text{engineer}(x) \wedge \text{contractor}(x))$

G: $\exists x(\text{engineer}(x) \wedge \sim \text{dependable}(x))$

F1: $\forall x(\text{dancer}(x) \rightarrow \text{graceful}(x))$

F2: $\text{student}(\text{Ayesha})$, F3: $\text{dancer}(\text{Ayesha})$

G: $\exists x(\text{student}(x) \wedge \text{graceful}(x))$

Example to be worked out in class

Every passenger is either in first class or second class. Each passenger is in second class if and only if the passenger is not wealthy. Some passengers are wealthy. Not all passengers are wealthy. Therefore some passengers are in second class.

Thank you